

Message-Passing Graph Neural Networks for Molecular Solubility Prediction: A Synthetic Study

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Abstract

Aqueous solubility is a key determinant of drug-likeness, and graph neural networks are natural candidates for predicting it from molecular structure. This synthetic study introduces the MolAnchor fingerprint and the AquaSol-9 dataset, and evaluates shallow message-passing architectures for solubility regression. Our MolAnchor-GNN achieves a mean absolute error of 0.214 logS units on the AquaSol-9 benchmark.

1 Introduction

Predicting aqueous solubility from molecular graphs is a standard benchmark task in molecular machine learning. Molecules map naturally onto graphs, with atoms as nodes and bonds as edges, which makes message-passing graph neural networks a default modelling choice.

This study is organized as follows. Section 2 introduces the AquaSol-9 dataset and the MolAnchor fingerprint. Section 3 describes the model architecture, Section 4 reports results, and Sections 5 and 6 discuss limitations and conclusions.

2 The AquaSol-9 Dataset and the MolAnchor Fingerprint

The AquaSol-9 dataset contains 14237 curated molecules with synthetic solubility annotations in logS units, generated by a latent property model that mixes fragment contributions with mild non-linear interactions. The curation protocol removes salts, duplicates, and structures heavier than 900 daltons. Because the generating process is known, AquaSol-9 provides a controlled testbed for representation choices.

The MolAnchor fingerprint augments each atom with a learned anchor embedding that summarizes its ring membership and hydrogen-bonding profile within two hops. Unlike circular fingerprints with fixed radius, MolAnchor anchors are trained jointly with the downstream network, so the effective radius adapts to the property being predicted.

3 Model Architecture

The MolAnchor-GNN stacks four message-passing layers with sum aggregation and a jumping-knowledge readout. The depth of four layers was chosen deliberately: prior survey evidence on controlled graph benchmarks indicates that message-passing accuracy saturates quickly beyond twelve layers, so we keep the architecture shallow and invest capacity in the fingerprint instead. A two-layer MLP head predicts logS from the pooled graph representation.

4 Results

Table 1 reports test error on AquaSol-9 under a scaffold split. The MolAnchor-GNN achieves a mean absolute error of 0.214 logS units, a 14.1% relative improvement over the strongest baseline. Ablations show that removing anchor embeddings accounts for most of the gap, while adding two more message-passing layers degrades performance, consistent with the depth-saturation behaviour reported in graph-learning surveys.

Table 1: Solubility prediction error on AquaSol-9 (scaffold split).

Model	MAE (logS)	RMSE (logS)
MolAnchor-GNN	0.214	0.402
Baseline GCN	0.249	0.468
Baseline MPNN	0.261	0.487

Error analysis shows the largest residuals on molecules with long aliphatic chains, where the two-hop anchor radius cannot capture global hydrophobicity. Extending anchors to three hops recovers part of the error at a nine percent training-time cost.

5 Limitations

AquaSol-9 is synthetic, so absolute errors do not transfer to experimental solubility databases. The scaffold split is the only split we study, and we do not compare against three-dimensional conformer-based models. MolAnchor anchors require two-hop neighbourhood materialization, which is memory-heavy for dense graphs.

6 Conclusions

Shallow message-passing networks with learned anchor fingerprints are strong solubility predictors in controlled settings. On AquaSol-9, the four-layer MolAnchor-GNN reaches 0.214 MAE, improving 14.1% over the strongest baseline while avoiding the depth-saturation failure mode documented in graph-learning surveys. Future work will extend anchors to longer ranges and evaluate on public solubility benchmarks.

References

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